Hector Lugo III

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Education

The University of Texas Rio Grande Valley Expected Dec. 2026
M.S. in Computer Science (GPA: 3.79/4.00) | B.S. in Computer Science (Aug. 2020 – May 2024)

Experience

AI/ML Research Intern May 2026 – Present
Air Force Research Laboratory (AFRL), Rome, NY
- Research how vision-language models understand multi-camera video, comparing centralized vs. decentralized ways of combining camera feeds on a 1,000-question benchmark; the best centralized method reached 61.8% vs. 55.7%.
- Built and open-sourced the end-to-end evaluation pipeline on a multi-GPU cluster; analyze failure cases to pinpoint where each approach breaks down, informing the design of stronger multi-camera systems.

Machine Learning Engineer Intern May 2025 – June 2026
Idaho National Laboratory (INL)
- Engineered a ResNet autoencoder to denoise smart-grid sensor signals for anomaly detection.
- Accelerated model training by over 93% (from 30 days to 1–2 days) with signal preprocessing and parallel computing.

Graduate Teaching Assistant Jan. 2024 – Present
The University of Texas Rio Grande Valley (UTRGV)
- Teach CS 1101 (Intro to CS) and Digital Image Processing, handling lectures, labs, and grading.
- Mentor 40+ students on programming fundamentals and image-processing coursework.

Research Assistant Jan. 2023 – Present
Machine Intelligence Lab, UTRGV
- Conduct Vision-Language Model (VLM) and Vision-Language-Action (VLA) research for humanoid locomotion.
- Build Isaac Sim environments and train reinforcement learning policies for Unitree Go2 quadruped navigation.
- Design autoencoders that compress robot state inputs, cutting compute cost and speeding up training.

Summer Research Intern June 2023 – Aug. 2023
University of California, Riverside
- Applied spectral clustering to U.S. railway accident data, identifying critical spatial safety zones.
- Strengthened predictive risk assessment for railway safety planning (published at ASME JRC 2024).

Projects

Multi-Agent Neuro-Symbolic Video Search | *Temporal Logic, VLMs* | [code] In Progress
- Designing a multi-agent system that searches asynchronous video streams from distributed sources: per-stream agents evaluate local temporal-logic queries, and a central orchestrator fuses results against a global neuro-symbolic specification (e.g., tracking an entity across disjoint camera networks).
- Extending retrieval into automated response: matches trigger analyst alerts, focused collection, or follow-up queries.

Real-Time Object Detection & Tracking | *YOLOv8, SORT, OpenCV* | [code] 2025
- Built a real-time multi-object detection and tracking pipeline: YOLOv8 detections feed a SORT tracker (Kalman filter + Hungarian matching) that keeps unique per-object IDs across frames.
- Supports class-aware tracking across all 80 COCO classes, with FP16 GPU inference, trajectory trails, and a configurable CLI for webcam or video input.

Personalized Chemotherapy Dosing | *Decision Transformer, Offline RL* June 2024 – July 2024
- Developed a Decision Transformer that learns personalized chemotherapy dosing from offline patient data.
- Framed treatment planning as offline RL, conditioning each dose on evolving patient state (published at ICICT 2025).

Publications

- **IEEE ICMLA 2026** — Vision–Language Guided Quadruped Navigation with Reinforcement Learning Control (accepted).
- **ICICT 2025** — Decision Transformers for adaptive chemotherapy dosing via offline RL. [paper]
- **ICICT 2025** — offline reinforcement learning for safe and effective smart-grid control. [paper]
- **ASME JRC 2024** — spatial safety analysis of U.S. railway accidents via spectral clustering. [paper]

Technical Skills

Languages: Python, C++, Java, SQL
Frameworks & Tools: PyTorch, OpenCV, NumPy, Pandas, Git, Docker, Linux, HPC (Slurm), FastAPI, Flask
Robotics & Edge: Isaac Lab, Isaac Sim, MuJoCo, LocoMujoco, ROS, Unitree Go2, NVIDIA Jetson Nano
Concepts: Reinforcement Learning, VLMs/VLAs, Multi-Agent Systems, Multi-Camera Perception, Computer Vision